University Mohammed VI Polytechnic

College of Computing

From Statistical Description to Sequence Modelling:

A Feasibility Study of Machine Learning for Eye-Tracking on Electrocardiogram Interpretation

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1 Introduction

Expert clinicians read an electrocardiogram (ECG) differently from novices, and eye-tracking makes that difference measurable. A decade of research has established it: experienced practitioners produce fewer fixations, shorter dwell times and much faster times-to-first-fixation on the diagnostically critical leads, while novices scan in a dispersed, exhaustive, rule-following manner [1, 2, 4, 5]. Gaze carries a signature of diagnostic competence.

The way that signature has been established is statistical. A study collects fixation counts per Area of Interest (AOI), groups participants by expertise, and applies non-parametric tests to reject the null hypothesis that the groups are the same. This is rigorous and it has produced real knowledge — the rhythm strip is fixated longest and lead I least [1]; interpretation accuracy climbs from 52% among medical students to 98% among consultants [2]. But it is limited by construction to *group-level*, *univariate* statements. It says nothing about an individual scanpath, it discards the temporal order of the fixations, and it produces no artefact that can be deployed.

This report documents the work we carried out at the UM6P College of Computing on exactly that gap. Our question was one of *feasibility*: can machine learning replace — or at least complement — the statistical study of ECG gaze behaviour with models that treat the scanpath as a sequential object? Section 2 presents the host institution and where our work sits inside it, Section 3 reviews the research done in the space, and Section 4 describes our work: the formulations we abandoned and why, the data preparation, the Hidden Markov Model (HMM) we chose, and the experimental and evaluation setups. Section 5 reports the results and examines what each metric does and does not establish; Section 6 sets out what a serious continuation requires and Section 7 concludes.

2 Presentation of the host institution

2.1 Domain of activity

Université Mohammed VI Polytechnique (UM6P) is a private non-profit research university created in 2013 and inaugurated in January 2017, established under the OCP Foundation. Its main campus is in Ben Guerir, inside the Mohammed VI Green City, with further campuses in Rabat and Laayoune and international hubs in Paris, New York and Montreal. Its stated mission is applied research and innovation contributing to regional economic and human development, which in practice means that research projects are expected to connect to a concrete problem rather than to remain purely academic — a point that turned out to matter a great deal for our own work, as Section 6 explains.

The **College of Computing** is the entity within UM6P that groups the university's computing activity. Its domain covers artificial intelligence, cybersecurity, cloud and software systems, and data science, spanning teaching, research and technology transfer. It operates on the Ben Guerir campus and, more recently, on the Rabat campus. Our project sits in its AI-and-health intersection: applying machine learning to physiological and behavioural signals — here, eye movements recorded during ECG interpretation.

2.2 Organisational structure

The College of Computing federates teaching and research under one roof rather than operating as a single department. The entity we worked within is the *School of Computer Science* (UM6P-CS), launched in 2020, which delivers the engineering and graduate programmes in computer science and hosts the faculty-led research groups; our project belongs to one of those groups. The college also draws on university-wide research infrastructure, notably the African Supercomputing Center and its Toubkal supercomputer, inaugurated in 2021. Figure 1 places our project inside this structure.

2.3 Professions present

The population of the college is a mix rather than a single profession, and the mix is what makes a project like ours possible. *Faculty members* define research directions, supervise and teach — our

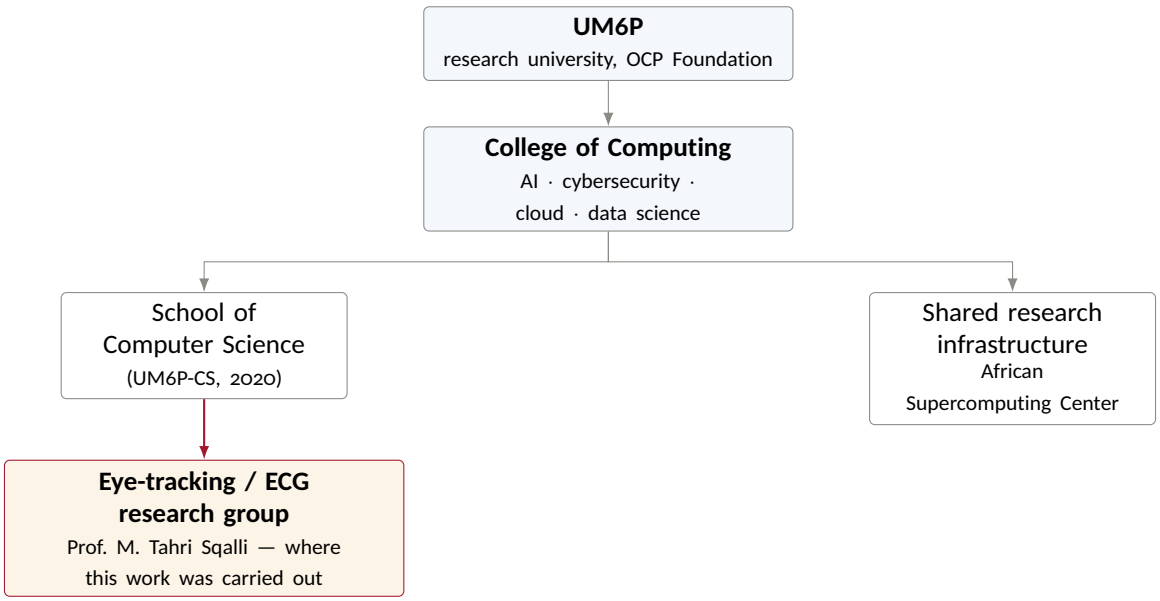


Figure 1: Organisational position of the project within UM6P and its College of Computing.

own supervision came from this group; *post-doctoral researchers and PhD candidates* carry the long-horizon research; *research and software engineers* build and maintain the experimental platforms, data pipelines and computing environments; *data scientists and machine learning engineers* turn collected data into models; *engineering and master's students* — our own position — contribute to projects on a fixed horizon under supervision; and *administrative and programme staff* handle admissions, partnerships and research administration. What shaped our work most directly, though, was access rather than headcount: our supervisor, Professor Mohammed Tahri Sqalli, holds the raw recordings behind an ECG eye-tracking corpus with few equivalents anywhere, whose published processed version is what other groups in this area reuse, and we worked from the unprocessed one.

3 Scientific context and problem statement

3.1 Research done in the space

We spent the first phase of the project reading, and the shape of the literature is what pushed us towards sequence modelling.

The statistical layer. The foundational study is Wood et al. [4], the first to apply the visual-search paradigm to ECG interpretation: ten final-year medical students against ten consultant emergency physicians on sixteen ECGs. Experts were roughly twice as accurate, twice as fast and one-and-a-half times more confident, and located critical leads considerably earlier; clinical history had no significant effect on abnormality detection. Sqalli et al. then built the corpus we use. Their first paper [1] exposed sixteen medical students to ten ECGs for 30 seconds each, deliberately not optimising for accuracy but for the *visual scanning dynamics* behind a diagnosis, and established the rhythm-strip/lead-I asymmetry. Their second [2] extended this to 63 practitioners across five expertise categories and confirmed that both accuracy ($p = .02$) and scanning efficiency ($p = .03$) track expertise. The 2025 systematic review [5] reads this through the dual-process model — novices reason analytically, experts recognise patterns — and proposes a concrete engineering path: time to first fixation on key leads, dwell time, revisitation frequency, scanpath directness. We took that as our list of candidate features.

The machine-learning layer. Attempts to learn from this data are recent and uneven. Bokhari et al. [6] applied four neuromorphic architectures (SNN, SCNN, RSNN and a spiking convolutional LSTM) with Random Forest feature selection, reporting above 90% expertise-classification accuracy and 95.6% for the best model, on $12{,}588$ training and $3{,}147$ test samples. We read this carefully and

came away unable to reconstruct where those samples came from, given that the underlying corpus contains 630 interpretations. That is not a small objection: if samples are individual fixations drawn from the same participant into both train and test, the reported accuracy measures participant identity, not expertise. A more transparent effort [7] trains Random Forest, SVM and KNN on the same lineage of data and reports 84% accuracy for Random Forest, with a feature-importance analysis and evidence of separable clusters. We took that paper as our reference point and replicated it — and the replication is what redirected the whole project, for reasons we set out in Section 4.1. The broader review of machine learning on eye-tracking [14] covers 31 studies, mostly radiology, mostly CNNs (65%) on classification tasks (49%) — which told us that the dominant recipe in the field is designed for *images* and does not transfer to a short symbolic sequence.

The sequence layer, where we found our angle. Two lines of work convinced us. Davies et al. [8, 9] grouped participants by *outcome* (correct vs. incorrect interpreters) rather than expertise, analysed scanpaths with n-grams, found bigrams dominant, modelled gaze with first-order Markov chains, and compared groups with the Jensen–Shannon and Hellinger distances. They found significant differences in transition behaviour: *the order in which you read the ECG influences whether you get it right.* The second is Haji-Abolhassani and Clark's inverse Yarbus process [10], reached through DeepScan [11]: an HMM whose hidden states are attention targets initialised by k-means and whose observations are fixation positions, trained one per task, classifying a new scanpath by maximum posterior likelihood. DeepScan is its semantic counterpart — CNN embeddings of fixation patches compared with Smith–Waterman alignment — and its survey of alignment methods (Needleman–Wunsch, ScanMatch [12]) is where our behavioural metric comes from. **To the best of our knowledge nobody has applied a hidden-state sequence model to ECG expertise; we also checked ICMI proceedings and found no ECG eye-tracking publication at all, the closest being a gaze-informed Vision Transformer for driving decisions [16]. That gap is what we decided to probe.**

3.2 The problematic

> *Is it feasible to replace — or at least to complement — the statistical study of ECG gaze behaviour with machine learning models that operate on the scanpath as a sequential object; and if so, what is the smallest honest model that demonstrates it, and what does the data actually allow us to claim?*

We were not trying to beat a benchmark, because there is none. We were trying to establish three things: that the data can be put into a form a sequence model can consume; that a model trained on that form learns something non-trivial rather than memorising a marginal; and that we can evaluate such a model honestly — which turned out to be the hardest of the three.

4 Our work and its place in the college

Our position was that of a student researcher inside the eye-tracking research group, working under supervision on a cycle of reading, experiment design, implementation and reporting. What follows is the work that cycle produced.

4.1 The formulation we abandoned, and why

Our first instinct was the obvious one: supervised classification of expertise from gaze features with tree ensembles, since the sample size forbids deep networks. We built XGBoost, Random Forest and decision-tree models on the published processed data and obtained results that were, as we noted at the time, ***too good to be true***. Chasing the discrepancy took an uncomfortable amount of time, and the answer was not a bug in the code: our training construction ignored the structure of the eye-tracking experiment, so rows belonging to the same participant–stimulus recording were scattered across train and test, and the model was recognising *who was looking*, not *how*.

A remark on the published results we replicated. Because our own numbers looked wrong, we went back to [7] and replicated it directly — same family of models (Random Forest, SVM, KNN), same lineage of data — and we obtained accuracies in the same range as the published ones. That agreement was not reassuring, because after investigating how the data was fed to the models we concluded that the training construction itself is unsound: the samples used for training and for testing are not independent, so the reported accuracy is in large part a measure of overfitting to participant identity rather than of expertise recognition. **We therefore believe that results obtained this way can support false conclusions**, however carefully the classifier itself is tuned. The second problem is structural rather than procedural: a feature-vector classifier flattens each recording into a bag of summary statistics — fixation counts, mean durations, revisitation rates — and in doing so *discards the sequential nature of the data entirely*. Two participants who fixate the same regions for the same total time in completely different orders are, to such a model, the same participant. Yet the order is precisely what the literature says carries expertise [8]. We are not claiming the published conclusions are necessarily wrong; we are saying that this experimental construction cannot establish them.

Both findings fixed the direction of the rest of the work. The first fixed a permanent rule: **the unit of observation is the participant–stimulus pair**, never the fixation, and any split must be made at that level or above. The second told us to stop flattening. What we needed was a model class that (i) **consumes the scanpath *as a sequence*, so that order is part of the signal rather than something averaged away**, and (ii) **is estimable from a few hundred sequences rather than requiring the tens of thousands of samples a recurrent or transformer model would demand — in other words, can be trained from a limited amount of data.** Those two constraints together point at a small family of models, and Hidden Markov Models sit at its centre — which is how they entered this project.

4.2 The formulation we adopted

Let an ECG image be partitioned into M discrete Areas of Interest $\mathcal{A} = \{a_0, \ldots, a_{M-1}\}$. One *scanpath* is the ordered sequence of AOIs visited by one participant on one stimulus, $O = (o_1, \ldots, o_T)$ with $o_t \in \mathcal{A}$, of variable length T. The corpus is a set of 630 such sequences. Two questions can be asked of this object, and they are not the same question:

(A) **Discrimination.** Given expertise-conditioned models λ_E and λ_N, classify an unseen scanpath by the likelihood ratio $\log P(O \mid \lambda_E) \gtrless \log P(O \mid \lambda_N)$.

(B) **Generation.** Given one model λ fitted to the whole population, sample synthetic scanpaths $\tilde{O} \sim P(\cdot \mid \lambda)$ that are behaviourally indistinguishable from human ones.

We chose (B) first, deliberately. (A) is the more useful end product, but it requires splitting an already small corpus into expertise strata, and it can only be trusted once we know the model class captures scanpath structure at all. (B) is the feasibility test for (A): if a generative model cannot produce plausible scanpaths, the likelihood ratios in (A) are meaningless anyway.

We should state the limitation of (B) up front, because we saw it while implementing and it never went away. **A model of this form encodes only the distribution that generated the scanpaths, so the generated scanpath *does not depend on which ECG is being read*: the semantics of the stimulus enters nowhere.** We built it anyway, on the grounds that a clean negative characterisation is worth more than an unbuilt idea.

4.3 Data and preprocessing

We work on the ECG eye-tracking corpus of Sqalli et al.: 63 medical practitioners from five expertise categories — medical students, cardiology nurses, technicians, fellows and consultants — each interpreting ten 12-lead ECGs, giving 630 interpretations. Recording used a Tobii Pro X2-60 at 60 Hz with iMotions 8.1, each ECG shown for at most 30 seconds. The ten stimuli span normal sinus rhythm, atrial fibrillation, atrial flutter, complete heart block, left bundle branch block, ST-

elevation MI, ventricular paced rhythm, ventricular tachycardia, Wolff–Parkinson–White syndrome and hyperkalemia.

One distinction matters. The publicly available version of this corpus [3], which the studies of Section 3 work from, is already processed; we worked instead from the unprocessed recordings, which retain the full gaze-sample stream and the event records marking each trial. That is what makes the sequential formulation possible at all — the scanpaths below are reconstructed from the raw stream rather than read off a table of summary features — and it means the preprocessing that follows is ours, as is any defect in it.

The raw export is not a table of fixations: it interleaves gaze samples with event records (`StartSlide`, `StartMedia`, mouse events), and one fixation spans several consecutive samples. Our pipeline (Figure 2) drops every row that is not a valid fixation; de-duplicates on the triple (`source_file`, `SourceStimuliName`, `Fixation Index`) so that one row means one fixation; verifies that the Fixation Index is monotone within every participant–stimulus group — it is, for all 630 — and uses it as the temporal ordering variable; and assigns every fixation to a cell of a uniform $k \times k$ grid laid over the ECG image, producing the discrete symbol `State`.

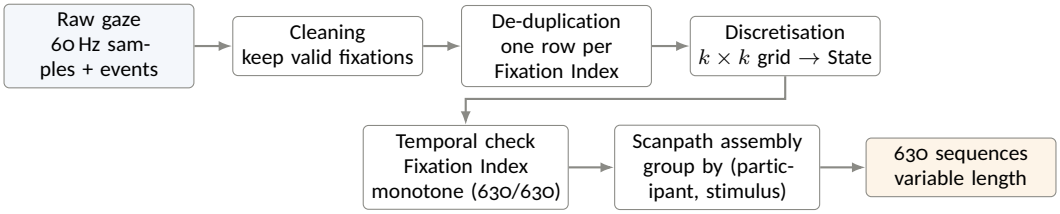


Figure 2: Preprocessing pipeline turning raw eye-tracking recordings into scanpath-level sequences.

Three modelling choices were made here, all in the direction of *not touching the data*. Scanpaths keep their natural variable lengths, because an HMM supports them natively and padding would falsify the recorded behaviour. No length standardisation is applied. And self-transitions — consecutive fixations in the same AOI, e.g. $[3, 3, 3, 0, 0, 1]$ — are preserved, so the model can learn persistence within a region as well as movement between regions. Persistence turns out to dominate everything else.

We report one inconsistency noticed while writing this up. The resulting scanpaths have a mean length of 233 fixations ($\sigma = 97$, median 251, range 6–424), $147{,}635$ fixations in total. Over a 30 second exposure that is roughly 7.8 fixations per second, well above the 2–4 Hz human oculomotor behaviour permits. We therefore believe the de-duplication step still lets some sample-level rows through for part of the corpus. This does not invalidate the transition structure we report — oversampling a fixation inflates the diagonal uniformly — but the absolute self-transition probabilities below should be read as an upper bound, and this is the first thing we would re-audit.

4.4 The method: Hidden Markov Models

An HMM is specified by $\lambda = (A, B, \pi)$ over N hidden states $S = \{s_1, \ldots, s_N\}$ and M observation symbols: the transition matrix $A = [a_{ij}]$ with $a_{ij} = P(q_{t+1} = s_j \mid q_t = s_i)$; the emission matrix $B = [b_j(k)]$ with $b_j(k) = P(o_t = v_k \mid q_t = s_j)$; and the initial distribution $\pi_i = P(q_1 = s_i)$. The joint probability of a state path and an observation sequence is

$$P(O, Q \mid \lambda) = \pi_{q_1} b_{q_1}(o_1) \prod_{t=2}^{T} a_{q_{t-1} q_t} b_{q_t}(o_t), \qquad P(O \mid \lambda) = \sum_{Q} P(O, Q \mid \lambda).$$

Following Rabiner [13], three problems must be solved, and we use all three: **evaluation** of $P(O \mid \lambda)$ by the forward algorithm, which gives us a scoring function on unseen scanpaths; **decoding** of the most likely hidden path by Viterbi, which would let us interpret the latent variable; and **training**, the entry point without which there is no model.

Training by Baum–Welch. There is no analytical solution to the training problem and no procedure that finds a global optimum. What exists is an iterative scheme reaching a local maximum: the Baum–Welch algorithm, which is Expectation–Maximisation specialised to this model. Define $\xi_t(i,j) = P(q_t = s_i, q_{t+1} = s_j \mid O, \lambda)$ and $\gamma_t(i) = \sum_j \xi_t(i,j)$, computed from the forward and backward variables. $\gamma_t(i)$ is the posterior probability of occupying s_i at time t, so $\sum_t \gamma_t(i)$ is the expected number of visits to s_i and $\sum_t \xi_t(i,j)$ the expected number of $s_i \to s_j$ transitions. Re-estimation is then event counting under those expectations:

$$\bar{\pi}_i = \gamma_1(i), \qquad \bar{a}_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}, \qquad \bar{b}_j(k) = \frac{\sum_{t\,:\,o_t = v_k} \gamma_t(j)}{\sum_{t=1}^{T} \gamma_t(j)}.$$

Baum proved that either $\bar{\lambda} = \lambda$, in which case λ is a critical point of the likelihood, or $P(O \mid \bar{\lambda}) > P(O \mid \lambda)$ — the new model is strictly more likely to have produced the data — so iterating converges monotonically to a maximum-likelihood estimate. The E-step is the evaluation of Baum's auxiliary function $Q(\lambda, \bar{\lambda})$ and the M-step its maximisation; equivalently the whole thing is a constrained optimisation of $P(O \mid \lambda)$ subject to A, B and π remaining stochastic. Two consequences matter in practice. Convergence is to a *local* maximum only, and the likelihood surface here is riddled with them, which is why we run multiple random restarts. And because re-estimation is pure expected counting, states that are never meaningfully occupied simply do not move: the model degenerates quietly instead of failing loudly. We saw exactly that.

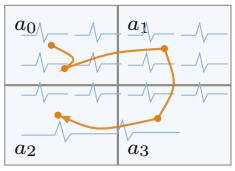
Generation. The same parameters can be used to generate new observation sequences by running the model forward. This generative process follows Algorithm 1.

Algorithm 1 Generating an observation sequence from an HMM

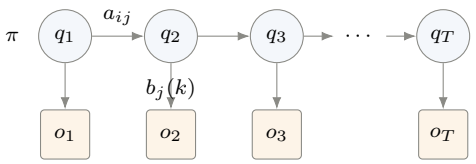
Input: HMM parameters $\lambda = (A, B, \pi)$, sequence length T
Output: observation sequence $O = (o_1, o_2, \ldots, o_T)$

1. Sample the initial state: $q_1 \sim \pi$
2. Emit the first observation: $o_1 \sim b_{q_1}(\cdot)$
3. For $t = 2$ to T:
    a. Sample the next state: $q_t \sim a_{q_{t-1},\cdot}$
    b. Emit the observation: $o_t \sim b_{q_t}(\cdot)$
4. Return $O = (o_1, o_2, \ldots, o_T)$

This dual interpretation — λ as both a model of how the observations were produced and as a machine that generates them — is what enables formulation (B). Figure 3 shows the discretisation alongside the model structure.



(a) $k = 2$ discretisation



hidden: strategy / focus (uninterpreted)
observed: AOI symbol $o_t \in \{a_0, \ldots, a_3\}$

(b) the model $\lambda = (A, B, \pi)$

Figure 3: Left: the 2×2 grid over a schematic 12-lead ECG with one illustrative scanpath; the sequence of visited cells, self-transitions included, is the observation sequence. Right: the HMM as a graphical model. We deliberately do not pre-assign a meaning to the hidden states.

4.5 Experimental setup

We kept the experiment small, since the point was feasibility. All experiments use a single population-level HMM (no expertise conditioning yet) with the configuration of Table 1.

Table 1: Experimental configuration.

Item	Value
Spatial discretisation	uniform $k \times k$ grid, $k = 2 \Rightarrow M = 4$ AOIs
Hidden states	$N = 4$
Emission family	categorical (discrete symbols)
Estimation	Baum–Welch (EM), max. 100 iterations, tolerance 10^{-4}
Random restarts	10 initialisations, best likelihood retained
Seed	42
Split	$80/20$ at the *scanpath* level: 504 train / 126 test
Train volume	$117{,}660$ fixations; test $29{,}975$
Test coverage	57 distinct participants over all 10 stimuli

Two points warrant clarification. First, the data split is performed at the scanpath level, which follows directly from the data leakage issue discussed earlier. However, this means a single participant could still appear in both the training and test sets through different stimuli. Before making any claim about expertise classification, we must therefore implement a stricter split that ensures participants are not shared across sets. Second, our choice of $k = 2$ is coarse: dividing a 12-lead display into only four quadrants is far from a detailed lead-level AOI map. We selected this resolution because it allows a 4-state HMM to be reliably estimated from 504 sequences. Our plan was to increase k gradually in later experiments; as the results show, this became the primary direction for future work.

4.6 Evaluation setup and the meaning of the metrics

Evaluating a generative model of behaviour is genuinely hard, and the design of this protocol took longer than the modelling. A likelihood alone is not enough — a model can assign high likelihood to held-out data while producing synthetic scanpaths no human would produce. So we built three branches that fail in different ways (Figure 4).

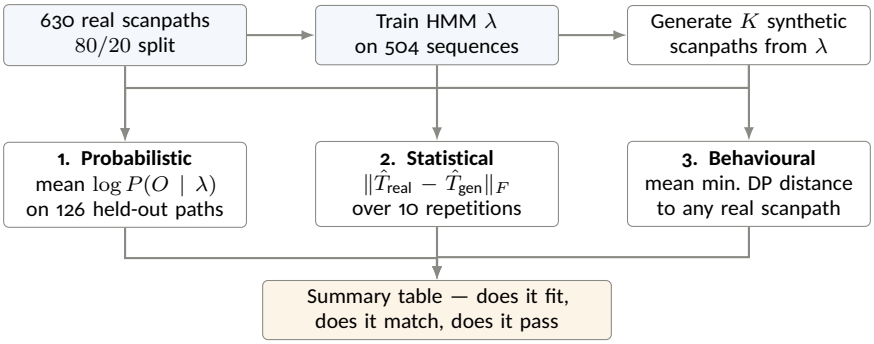


Figure 4: The three-branch evaluation protocol as we designed it. Each branch answers a different question and can fail while the others pass, which is why we ran all three; Section 5.1 examines what each one turned out to measure.

1. Probabilistic — mean held-out log-likelihood. We compute $\log P(O \mid \lambda)$ by the forward algorithm on each of the 126 test scanpaths and normalise by length, $\ell(O) = \log P(O \mid \lambda)/T$, because raw log-likelihoods are dominated by sequence length and are not comparable across scanpaths. The normalised quantity is interpretable: $\exp(-\ell)$ is the model's *perplexity per fixation*, the effective number of AOIs the model hesitates between at each step. It answers *does the model fit unseen human data?* and its failure mode is overfitting. Its reference points are the

uniform baseline $\log(1/M) = -1.386$ (a model that knows nothing) and the unigram baseline $\sum_k p_k \log p_k = -1.350$ (a model that knows only how often each AOI is visited, and nothing about order). Beating the unigram is the minimum bar for claiming that sequential structure was learned at all.

2. Statistical — Frobenius distance between transition matrices. From the real test scanpaths and from a batch of generated ones we estimate first-order AOI transition matrices $\hat{T}_{ij} = c_{ij} / \sum_j c_{ij}$ by maximum likelihood and compare them with $\|\hat{T}_{\text{real}} - \hat{T}_{\text{gen}}\|_F$. This asks *does the model reproduce the aggregate dynamics?* We were aware of its weakness — regenerating a fresh batch gives a different value — so we repeat generation ten times and report mean and standard deviation. We considered the Jensen–Shannon and Hellinger distances used by Davies et al. [8, 9] and regard them as the more principled choice, being distances between *distributions* rather than between matrices of numbers; Frobenius was the pragmatic first pass.

One detail of the design is important. $\hat{T}_{\text{real}}$ is estimated on the 126 *held-out* scanpaths, not on the 504 the model was trained on. The branch therefore asks whether the transition structure learned from the training participants also describes participants the model never saw, which is a generalisation test and can fail. What it needs in addition is a reference for how small a distance counts as small; we return to this in Section 5.1.

3. Behavioural — minimum dynamic-programming distance. This is the branch we care about most and the one we designed ourselves, by analogy with sequence alignment in genomics and with the scanpath-comparison literature [12, 11]. For each generated scanpath $\tilde{O}$ we compute an edit (Levenshtein-type) distance by dynamic programming against *every* real scanpath and keep the minimum, normalised to a similarity in $[0, 1]$:

$$\text{sim}(\tilde{O}) = \max_{O \in \mathcal{D}} \left(1 - \frac{d(\tilde{O}, O)}{\max(|\tilde{O}|, |O|)} \right), \qquad \text{score} = \frac{1}{K} \sum_{k=1}^{K} \text{sim}(\tilde{O}_k).$$

The question it asks is deliberately concrete: *could this generated scanpath plausibly have been produced by a human?* Unlike the transition-matrix comparison it is sensitive to order and to local structure, not only to aggregate frequencies. Its cost is quadratic in sequence length and linear in corpus size, which is why K is modest.

Calibration was the main challenge with this approach, and it took us time to recognise it. A similarity score such as 0.54 is meaningless by itself, because the scale has no fixed reference points. It has no upper anchor, because even two *real* human scanpaths never match perfectly: the best possible score a generator can achieve is whatever real humans score against each other. This is the *ceiling*. It has no lower anchor either, because with only four symbols and long repetitive sequences, even unrelated scanpaths match somewhat by chance. The worst-case score therefore comes from a generator that produces sequences with no meaningful structure. This is the *floor*. We compute both anchors alongside our model: real scanpaths compared to other real scanpaths give the ceiling, and random draws from the marginal AOI distribution, which ignore order, give the floor. Section 5.1 reports these values.

5 Results and analysis

Table 2 gathers the results of the three branches. We report them below, then turn to the estimated parameters themselves.

The model fits held-out human scanpaths. The mean per-fixation log-likelihood on the 126 unseen scanpaths is -0.494, against -1.350 for a unigram model and -1.386 for uniform guessing; in perplexity terms the model hesitates between 1.64 AOIs at each fixation where the unigram hesitates between 3.86. The distribution (Figure 6a) is unimodal and every held-out scanpath clears both baselines, so there is no subpopulation the model fails on.

Table 2: Evaluation results. The unigram baseline is the bar that must be cleared for sequential structure to have been learned.

Branch	Metric	Value
Probabilistic	mean per-fixation log-likelihood ℓ	-0.494 ± 0.209
	perplexity per fixation	1.638
	uniform / unigram baseline	-1.386 / -1.350
Statistical	$\|\hat{T}_{\text{real}} - \hat{T}_{\text{gen}}\|_F$	0.0135
	mean self-transition, real vs. generated	0.874 / 0.875
	entropy rate, real vs. generated (bits/fixation)	0.687 / 0.684
Behavioural	mean min. DP similarity (10 repetitions)	0.539 ± 0.005

The generated scanpaths reproduce the held-out transition structure. Figure 5 puts the empirical and synthetic transition matrices side by side with their absolute difference. No entry differs by more than 0.008 and the Frobenius distance is 0.0135. The marginal AOI distribution is reproduced to within about one percentage point per cell (Figure 6b), and two further checks agree: the mean self-transition probability is 0.874 in the real data against 0.875 in the generated, and the entropy rate 0.687 against 0.684 bits per fixation. The behavioural branch gives 0.539 ± 0.005 over ten independent generation batches; the small standard deviation confirms that repeating the generation removed the sampling noise we were worried about.

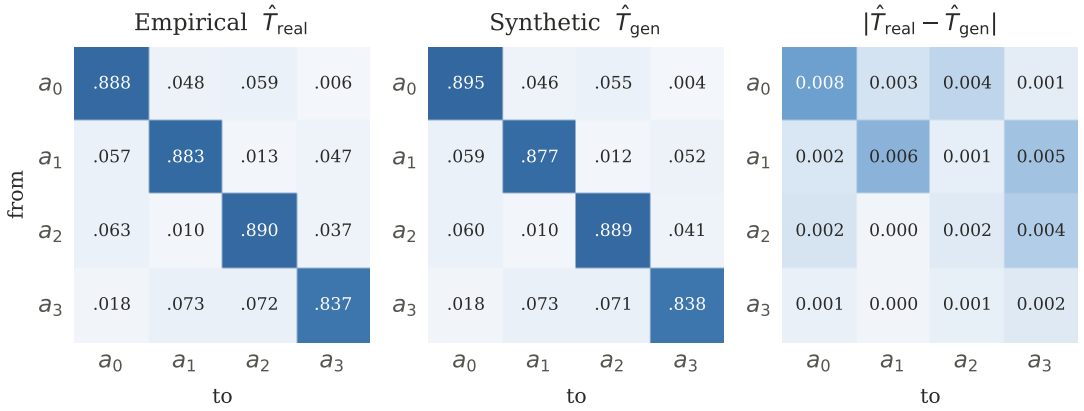
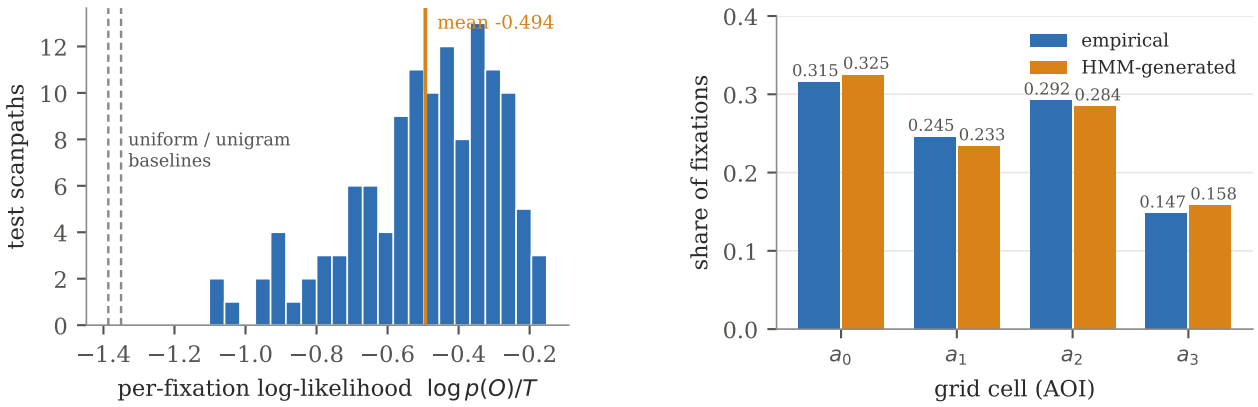


Figure 5: First-order AOI transition matrices estimated by maximum likelihood from the 126 held-out human scanpaths (left) and from the HMM-generated scanpaths (centre), with the absolute difference (right, on a $100\times$ finer colour scale). Both are overwhelmingly diagonal.

The learned parameters. Metrics summarise a model; they do not describe it. Inspecting the estimated parameters directly, we found the emission matrix (rows hidden states, columns AOIs)

$$B = \begin{pmatrix} 0.980 & 0.004 & 0.016 & 0.000 \\ 0.019 & 0.001 & 0.974 & 0.007 \\ 0.001 & 0.021 & 0.017 & 0.960 \\ 0.013 & 0.971 & 0.001 & 0.015 \end{pmatrix}$$

which is, to three decimals, a permutation matrix: each hidden state emits one AOI and essentially nothing else. The estimated hidden-state transition matrix carries 0.90–0.93 on its diagonal and the posterior state occupancy is $(0.313, 0.291, 0.148, 0.249)$, matching the AOI marginals. In other words, with $N = M = 4$ Baum–Welch converged to a solution in which the states align one-to-one with the observations. This is a property of the configuration rather than of the data: at this resolution there are exactly as many latent states as observable symbols, so there is nothing for the latent layer to group or separate, and the estimator has no reason to place the states anywhere else. It is the clearest single argument for the change of resolution proposed in Section 6.



(a) held-out log-likelihood **(b)** AOI occupancy

Figure 6: (a) Distribution of the per-fixation held-out log-likelihood over the 126 test scanpaths; every one clears both baselines. (b) Marginal AOI occupancy, empirical vs. generated.

5.1 What each metric does and does not establish

Three good-looking numbers are not three independent confirmations, and going back through the branches to ask what each one is actually sensitive to is, we think, the most transferable part of this study. The governing principle is simple, and we had not applied it when we designed the protocol: *a metric carries information only if something could have failed it.*

The likelihood measures fit, against weak reference points. The value of -0.494 is a genuine statement about held-out data, and the model clearly does better than a generator that knows only the AOI frequencies. What it does not tell us is *which* regularity is being exploited, and the answer is largely the persistence of fixation: with a probability of 0.867 of remaining in the same quadrant, "the same AOI again" is by itself a strong prediction, and a perplexity of 1.64 is close to what that single number implies. A likelihood is a summary of how predictable the data is, not a certificate that the model found interesting structure.

The statistical branch tests generalisation, and needs a noise reference. Because $\hat{T}_{\text{real}}$ is estimated on held-out scanpaths, a large distance here would have meant that the transition structure learned on the training participants does not describe new ones. The branch could fail, and it does not. What was missing was a sense of scale: is 0.0135 small? We answer it by asking how much two samples of the *real* data differ from each other for no reason other than sampling. Drawing two bootstrap resamples of the 126 held-out scanpaths and comparing their transition matrices gives a mean distance of 0.035, with 90% of the values between 0.018 and 0.059. The model-to-real distance of 0.0135 is smaller than that. In other words the generated transition structure differs from the held-out human one by less than the human data differs from itself, so on this statistic the match is as close as the sample size allows. Two qualifications keep this in proportion. Sequences drawn from a fitted model vary less than resampled real data, which accounts for part of the gap. And the statistic is a first-order aggregate, pooled over all participants and all ten ECGs: it is blind to structure beyond one step, to differences between individuals, and to differences between expertise groups. Passing it establishes that one-step transition structure generalises, not that scanning behaviour as a whole is reproduced. Comparing the chains of two groups, as Davies et al. do with the Jensen–Shannon distance [8], is the version of this branch that carries a genuine null hypothesis.

The behavioural branch needs calibration to be readable at all. The similarity of 0.539 looked like a mid-scale number only because we had nothing to compare it with. Table 3 supplies the anchors defined in Section 4, computed against the same 126 held-out scanpaths.

Two conclusions follow. First, the meaningful range of scores is much smaller than the raw numbers suggest. A random generator that only knows the marginal AOI frequencies already achieves 0.424, because two long, repetitive sequences over four symbols will match by chance

Table 3: Calibration of the behavioural branch: minimum dynamic-programming similarity against the 126 held-out human scanpaths, for the model and for the two anchors.

	Generator of the compared sequences	Similarity
floor	independent draws from the marginal AOI distribution	0.424
model	**the trained HMM**	$\mathbf{0.535}$
ceiling	real human scanpaths against other real humans	0.566

quite often. So the actual range where a model can show its value runs from 0.424 to 0.566 — the latter being the best score any generator could reasonably achieve, since even real humans differ from each other by that amount. Our model scores 0.535, placing it closer to the ceiling than the floor, but not at the top.

Second, and more importantly for anyone using this method, a similarity score without these two reference points is meaningless. A value of 0.539 tells you nothing unless you also know that 0.424 is random chance and 0.566 is human-level performance.

Why the states matched the AOIs. We believe two factors caused this, and they reinforce each other. First, resolution: with $k = 2$ there are only four regions. This means most clinically meaningful eye movements — between leads or between waveform components — occur *within* a single AOI, so the model cannot see them. Second, the sample-level duplication we mentioned earlier inflates the diagonal transitions, making self-transitions dominate. Both factors have the same effect: they make the sequences appear much more repetitive and less structured than the actual behaviour really is.

5.2 Conclusion

In short, the pipeline works, but the model in its current configuration does not yet justify its use.

On the positive side, we can successfully convert the corpus into clean sequential data. The temporal ordering is reliable for all 630 recordings. The HMM trains stably and produces reproducible results. And our evaluation protocol gives interpretable numbers rather than a single opaque score. **This protocol is the most reusable contribution of our work: it can be applied directly to higher-order Markov chains, expertise-conditioned HMMs, or any other generative scanpath model.**

On the limiting side, two issues are clear. *First*, the model is stimulus-blind — it never sees which ECG is being read — so it cannot support any clinically meaningful claim. This is a limitation of the formulation, not of the parameter estimation. *Second*, the chosen resolution is too coarse for the behaviour we aimed to model, and the estimated parameters clearly show this.

6 Perspectives

Fix the immediate defects first. Re-audit the de-duplication so that one row is provably one fixation; sweep $k \in \{3, 4, 6\}$ and, better, replace the uniform grid with clinically meaningful AOIs — the twelve leads and the rhythm strip — or with data-driven regions from k-means or DBSCAN on the fixation coordinates as in [10, 9]; promote the baselines and anchors of Table 3 to a permanent part of the protocol, reported beside every model; and enforce a participant-disjoint split. With $M \gg N$ the states can no longer align one-to-one with the symbols, and the latent layer is given something to do. Then, and only then, is formulation (A) worth attempting: one HMM per expertise stratum, classification by likelihood ratio, with Jensen–Shannon or Hellinger distances between the learned chains as the descriptive counterpart.

The data is the binding constraint. 630 sequences from 63 participants is small for anything with hidden structure, and it is the reason the more ambitious results in the literature [6] are hard to trust. Richer data is worth more than a richer model here — more participants, more stimuli per participant, and recordings not capped at 30 seconds, since the cap truncates exactly the long deliberative scanning that distinguishes a struggling novice. Higher-order Markov chains ($n \geq 3$)

need this volume before they can be estimated at all.

Multiple modalities. The scanpath is only one of several channels the recordings already contain, or could contain: fixation durations (currently discarded — we model *where* a person looks, but not *how long*), pupil dilation as a measure of cognitive load, the ECG signal as a time series, and the ECG image itself. The direction we find most promising is to give the model the stimulus as well as the gaze, because that is the only way to remove the stimulus-blindness described above. We leave this as future work, and we would start from the HAIM framework of Soenksen et al. [15], which combines tabular records, time series, text and images in one modular pipeline and reports gains of 6–33% over single-modality models across twelve clinical tasks. Its modular design suits our case: each channel keeps its own pretrained feature extractor and the resulting embeddings are combined, so we would not have to train a large model from scratch. DeepScan [11] remains the template for the gaze side, though its CNN-over-fixation-patches recipe needs far more data than we have; a smaller pretrained embedding is the realistic substitute.

Saliency modelling. The natural reformulation is spatial rather than sequential: *where do expert cardiologists look when they read an ECG?* Answering it means building a saliency model over the ECG image conditioned on expertise, then comparing a student's empirical map against it. The two variants — predicting a saliency map, or predicting the scanpath itself — have different data requirements and evaluation traditions, so the choice should be made deliberately rather than drifted into.

Above all: fix a single industrial question. The strongest conclusion we draw from this phase is methodological, and it connects directly to the applied-research mission of the host institution described in Section 2. We noted while implementing the HMM that we could not name the real input/output of the model, and that uncertainty is visible in everything above: we optimised a generator without a consumer for what it generates. The next phase should begin by committing to *one* concrete question with an identified user and an identified decision. The modelling here is not blocked on algorithms; it is blocked on the absence of a fixed question.

7 Conclusion

We set out to test whether machine learning can take over from statistical description in ECG eye-tracking research. Our answer, from this phase, is a qualified yes with an important caveat about method. The data can be made sequential, an HMM can be trained and scored on it, and the protocol returns interpretable numbers. But at the resolution we used the estimated states aligned one-to-one with the observed AOIs, and our metrics could not have told us so — we found it by reading the parameters. That is what the study establishes most firmly, and it tells us exactly what the next iteration must change: finer and clinically grounded AOIs, anchored metrics, a participant-disjoint protocol, stimulus information in the model, gathering more data, and a single fixed industrial question to give all of it a purpose.

Beyond the technical outcome, the work taught us something about method that no reading could have. Three times we were saved not by a better algorithm but by looking again at how the data was produced: when our own leak turned into an accuracy too good to be true, when the replication of a published result reproduced its numbers and its flaw at the same time, and when the estimated emission matrix turned out to say more about the model than any of the three scores did. That is the habit we take away: read the data and the parameters, not only the metrics.

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Abstract

Eye-tracking research on electrocardiogram interpretation has, for a decade, been almost entirely descriptive: it counts fixations, measures dwell times, and confirms with non-parametric tests that experts look less and look better than novices. This report asks a different question — rather than *describing* gaze, can we *model* it? Using a corpus of 630 scanpaths from 63 medical practitioners, the study formulates the scanpath as a discrete sequence over Areas of Interest, trains a Hidden Markov Model on it, and evaluates it through a three-branch protocol — probabilistic, statistical and behavioural. The model reaches a per-fixation log-likelihood of -0.494 against a unigram baseline of -1.350, reproduces the empirical transition matrix to a Frobenius distance of 0.0135, and attains a mean dynamic-programming similarity of 0.539 ± 0.005 against held-out human scanpaths, which sits between a chance floor of 0.424 and a human–human ceiling of 0.566. Reading the estimated parameters shows that at this resolution the hidden states align one-to-one with the observed areas of interest, and that none of the three metrics could have revealed it — the most useful outcome of the study, and the one that dictates what the next iteration must change: finer and clinically grounded areas of interest, anchored metrics, a participant-disjoint protocol, stimulus information in the model, and a single fixed industrial question to give the work a purpose.

Keywords: eye-tracking; electrocardiogram interpretation; scanpath modelling; Hidden Markov Models; visual expertise; machine learning feasibility.